



KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

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FORMAT FOR STUDENT PROJECT PROPOSAL FOR THE 47th SERIES OF STUDENT PROJECT PROGRAMME

(Handwritten proposals will not be accepted, please fill all the details in this MS word file, insert images / diagrams wherever necessary. Convert to pdf file, get it approved from the project guide / head of the department and principal of your institution. Keep ready the scanned pdf file of 1) Declaration and Endorsement 2) details of processing fees made and fill-up the Google Form.

<https://forms.gle/mE8Q4pM2nwZQuHbi9>

1.	Name of the College: KLS GOGTE INSTITUTE OF TECHNOLOGY, BELGAVI
2.	Project Title: Accident detection and alert system using IOT
3.	Branch: Information Science and Engineering
4.	Theme (as per KSCST poster): Integrated Internet of things
5.	Name of project guide: 1. Name: Dr. Harish Kenchannavar Email id: harishhk@git.edu Contact No.: 7411475126
6.	Name of Team Members: Name: BASAVRAJ R DUGANAVAR USN No.: 2GI20IS006 Email id: basavarajdeepti@gmail.com Mobile No: 8618709461 Name: CHINMAY C USN No.: 2GI20IS009 Email id: chinmaychinchanikar@gmail.com Mobile No.: 7022096732



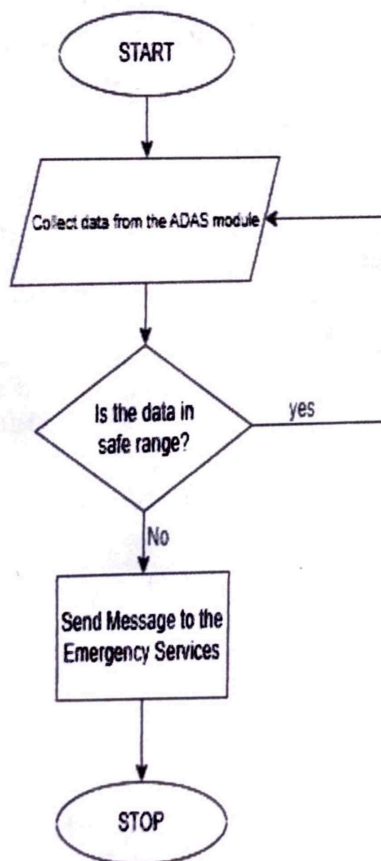
	Name: ADITYA VARDHAN USN No.: 2GI20IS003 Email id: adityavardhan5673@gmail.com Mobile No.: 6364433459	
	Name: NACHIKET M USN No.: 2GI20IS016 Email id: nachiketmagadum@gmail.com Mobile No.: 9611504055	
7.	Team Leader of the Project: Name: BASAVRAJ R DUGANAVAR USN No.: 2GI20IS006 Email id: basavarajdeepti@gmail.com Mobile No.: 8618709461	
8.	Processing Fee Details (Through Online Payment only): (processing fee of Rs. 1000/-) Please furnish the payment details in the format provided in the last page of the proposal.	
9.	Date of commencement of the Project: 20/11/2023	
10.	Probable date of completion of the project: 20/05/2024	
11.	Scope / Objectives of the project: ● Design a system that can detect real accidents and ignore non-emergency events ● Avoid false alarms and ensure that no real accidents go unreported. ● Reduce the time it takes for emergency personnel to arrive at the scene of an accident, thereby improving the chances of survival and speedy recovery for the injured person.	

12.

Methodology:

ALGORITHM:

- Accident Detection and Alert System using Arduino can prevent uncertain deaths by sending a message alert to a registered mobile number, providing the precise location of the accident through a Google map link.
- We Collect real-world accident data, including various scenarios and conditions. This will serve as foundation for training the system.
- To the collected data, machine learning algorithms(Clustering / Partitioning Algorithms) are applied to get the range of values that determine whether the disturbance was an accident or not
- The Accident Detection and Alert System checks the readings from the sensors and compares it with the analyzed range of values to distinguish between accidents and false alarms(Speed Bumps, Potholes, etc).
- If the disturbance is found to be an accident then an emergency message containing information such as GPS co-ordinates of the accident, time of the accident, Vehicle number is sent to the emergency services and the contact numbers mentioned by the user.
- Flowchart:



	Block Diagram: <pre> graph TD PS[POWER SUPPLY] --> AR[ARDUINO] AR --> LCD[16 X 2 LCD] AR --> GPS[GPS RECEIVER] AR --> GSM[GSM MODULE] AS[ACCELEROMETER SENSOR] --> AR </pre>
13.	Expected Outcome of the project: Reduced Emergency Response Time: By quickly detecting and accurately assessing the severity of accidents, the system facilitates a rapid response from emergency services, potentially reducing the time it takes for personnel to arrive at the accident scene. Improved Survival Rates: Swift and efficient communication with emergency services enhances the chances of timely medical intervention, contributing to improved survival rates and minimizing the impact of injuries sustained in accidents. Enhanced Reporting to Authorities: The system automates the reporting process to authorities, providing them with timely and accurate information about accidents. This can streamline emergency response coordination and enable better-informed decision-making by relevant agencies.
14.	Is the project proposed relevant to the Industry / Society or Institution? Yes / No: Yes(Societal)
15.	Can the product or process developed in the project be taken up for filing a Patent? Yes / No: Yes Prior Art search done? Yes/No: Yes

16.	Budget details (break-up details should be given): Note: KSCST will provide nominal grant support for carrying out the project by students if selected by the project selection committee. <table border="1"> <thead> <tr> <th>Budget</th> <th>Amount</th> </tr> </thead> <tbody> <tr> <td>a) Materials / Consumables (Please specify)</td> <td></td> </tr> <tr> <td>• SIM800 L(sim module)</td> <td>₹ 600/-</td> </tr> <tr> <td>• NEO GPS module</td> <td>₹ 1500/-</td> </tr> <tr> <td>• ARDUNIO mega</td> <td>₹1500/-</td> </tr> <tr> <td>• ADXL335/345</td> <td>₹1500/-</td> </tr> <tr> <td>• MQ-2 smoke sensor</td> <td>₹300/-</td> </tr> <tr> <td>• Heat sensor</td> <td>₹200/-</td> </tr> <tr> <td>• Tilt sensor</td> <td>₹ 300/-</td> </tr> <tr> <td>• Bread board</td> <td>₹200/-</td> </tr> <tr> <td>• LCD screen</td> <td>₹400/-</td> </tr> <tr> <td>• Wires</td> <td>₹400/-</td> </tr> <tr> <td>b) Labor (Describe) Cost of 3D Printing .</td> <td>2000.00/-</td> </tr> <tr> <td>c) Travel (Describe) Travel related to work, Traffic and Hospital Survey</td> <td>600.00/-</td> </tr> <tr> <td>e) Miscellaneous (Please specify) cost of Battery, Stationary</td> <td>500.00/-</td> </tr> <tr> <td>Total</td> <td>₹10,000.00/-</td> </tr> </tbody> </table>	Budget	Amount	a) Materials / Consumables (Please specify)		• SIM800 L(sim module)	₹ 600/-	• NEO GPS module	₹ 1500/-	• ARDUNIO mega	₹1500/-	• ADXL335/345	₹1500/-	• MQ-2 smoke sensor	₹300/-	• Heat sensor	₹200/-	• Tilt sensor	₹ 300/-	• Bread board	₹200/-	• LCD screen	₹400/-	• Wires	₹400/-	b) Labor (Describe) Cost of 3D Printing .	2000.00/-	c) Travel (Describe) Travel related to work, Traffic and Hospital Survey	600.00/-	e) Miscellaneous (Please specify) cost of Battery, Stationary	500.00/-	Total	₹10,000.00/-
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17.	Any other technical details (Please specify):																																
18.	SPP Coordinator (Identified by the college): Name: Prof. Shwetha Goudar Email id: dean_rnd@git.edu Contact No.: 0831-2498500																																

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